Affine Geometry and Paid Recursion for Subexponential SAT

A smaller certified exponential upper bound, an exact convolution kernel, and an updated review of the published formalizations

Research prepared for Vladimir Reshetnikov

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Abstract

We review ProveIt's twenty filed refinement packets and nine newly committed archives at `73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d`, with a closing review of their relocation through `7de9d09fec`, and incorporate the user's additional reference to Anthropic's now-public Lean development at revision `1a6eee293b9e00cb432eef6018a21f523623a9af`. For the common normalized Boolean tensor-contraction interface, we prove a new bound with time coefficient $30.2698729104\ldots$ and space coefficient $29.4875372869\ldots$ on the scale $s/\sqrt{\lceil \log_2 s \rceil}$. The strongest comparable submitted pair is $30.8910189204\ldots$ and $30.7539128321\ldots$.

The new ingredient is a four-plan contraction menu from the affine plane $\mathbb{F}_3^2$. Its uniform aggregation coefficient is exactly certified as $1/\sqrt{3}$, replacing $0.6021452276\ldots$ at nine groups. A short geometric proof bounds all operation supports, including intermediate table construction. A separate twelve-group certificate, punctured affine menus, and an explicit strict rational $9 \to 12 \to 8 \to 7 \to 4$ schedule pay for every recursive child entry. We also give an exact Toeplitz-convolution implementation for packet matrix actions and clarify why several tempting changes do not establish a smaller logarithmic order.

These are natural-language proofs with reproducible finite certificates, relative to stated inherited deformation, recovery, and packing interfaces. The generic result uses subexponential space. It improves the coefficient in an exponential upper bound, not the unrestricted SAT exponent order, and is not a measured full-solver speedup or a new formal verification of either primary manuscript.

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1 Result, comparison, and limits

Put $\lambda = \lceil \log_2 s \rceil$ and $\mathcal{E}(s) = s/\sqrt{\lambda}$, where s is the normalized number of Boolean wires. Under the interface in Section 4, this report establishes

$$T(s, \mathcal{L}) \le (s + \mathcal{L})^{O(1)} 2^{(30.2698729104411\ldots + o(1))\mathcal{E}(s)}, \tag{1}$$

$$S(s, \mathcal{L}) \le (s + \mathcal{L})^{O(1)} 2^{(29.4875372869212\ldots + o(1))\mathcal{E}(s)}. \tag{2}$$

Here $\mathcal{L}$ is the input description length. Convenient safe eventual coefficients are 30.270 and 29.488. Every recursively evaluated tensor entry is charged. The proof uses the same conservative matrix exponent 12/5 as the strongest comparable submitted tree certificate; using a newer matrix exponent is not the source of the gain.

Generic normalized interface	Time coefficient	Space coefficient
Strongest latest cached packet [2]	30.891019	30.753913
This report	**30.269873**	**29.487537**
Strongest polynomial-space packet [3]	46.184846	polynomial

The time coefficient decreases by about 2.01% and the space coefficient by about 4.12%. For the displayed asymptotic bound expressions,

$$\log_2 \frac{U_{\text{previous}}(s)}{U_{\text{new}}(s)} = (0.621146009925\ldots + o(1))\mathcal{E}(s).$$

This is an exponential separation of *certified upper-bound expressions*. It is not a lower bound on the previous algorithm's actual runtime. The new algorithm spends subexponential memory; it does not replace the polynomial-space branch when memory is constrained.

The mathematical change. Nine groups are the nine points of $\mathbb{F}_3^2$. Its four directions give four partitions into three parallel lines. Each partition specifies a full contraction plan. We choose the cheapest plan for the actual exact mode-list sizes. A geometric support bound gives aggregation coefficient $1/\sqrt{3} = 0.5773502691\ldots$, versus $0.6021452276\ldots$ for the previous nine-group menu. A twelve-group menu is separately certified by integer arithmetic. Puncturing one or two points from the affine plane supplies the later eight- and seven-group steps. This makes the entire recursion cheaper, not just its first operation.

What the result does not establish. The power of the logarithm is unchanged. The inherited density-aware SAT order remains

$$(n+m)^{O_k(1)}\, 2^{O_k\left(\frac{n}{\sqrt{\log n}} \min\{\log(2+m/n), \log\log n\}\right)}. \tag{3}$$

Thus the worst-case exponent is still of order $n \log\log n/\sqrt{\log n}$, and at fixed clause density it is $n/\sqrt{\log n}$. Small sizes are treated separately. The constant per wire cannot simply be placed before the number n of SAT variables: the reduction has its own expansion and arithmetic constants. We do not numerically dominate the unspecified constants in either distinct primary SAT theorem. Nor do we treat conditional $n/\log n$ proposals as completed algorithms. This is a new coefficient theorem and an exact kernel improvement, not a claimed breakthrough in the unrestricted logarithmic order.

2 Review of the recent repository state

2.1 Frozen source state and recent commits

The review is pinned to ProveIt `73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d`. At that revision, `Computability/SubexponentialSAT/Research` contains twenty filed refinement packets and one separate Claude manuscript. The OpenAI manuscript is vendored under `lib/openai-math/preprints`. Nine additional SAT archives are present in `docs/incoming`. All nine were safely extracted, passed ZIP CRC verification, and had their README, status, report, and relevant proof/code sources inspected. The selected mathematical dependencies were read in detail; this is not a line-by-line audit of every historical artifact.

The recent mathematical intake is summarized below; times are Pacific on October 26, 2026.

Commit	Time	Relevant change
`6d3adc4506`	19:03:44	Eight second-round packets placed.
`f42479bca2`, `f898a9fde4`, `fa6e904d10`	19:28–19:31	Five third-round packet arrivals.
`450252e9d8`	19:36:02	Those five packets placed; two replay scripts restored.
`b9941a7396`	20:11:06	Nine new SAT archives committed unchanged.
`73dfb5e5ab`	20:11:10	Mathematical intake snapshot after a merge.
`a7ae5ab55f`	20:15:51	Nine incoming packets filed; comparison README updated.
`7de9d09fec`	20:20:40	Public Anthropic formalization linked.

At the intake snapshot, the project README predates the nine new archives. Stopping at that earlier README would incorrectly use $35.354588\ldots$ as the cached target. The new $30.8910189\ldots$ target comes from the tree-certificate archive [2]. A closing review through `7de9d09fecd17de776d712835ecb2011205c531d` covers the two later commits: there are now twenty-nine filed packets, `docs/incoming` contains only its README, and the comparison README includes the new frontier. The archived source links in this report intentionally retain the original immutable ZIP snapshot. Exact Git-blob matching confirms the relocated research files against the original archive members. The provenance directory records complete commit identities, immutable paths, Git blob identifiers, and SHA-256 hashes for the reviewed corpus.

2.2 The nine new archives

The table abbreviates archive names; the complete names and source anchors are in `review/latest-frontier.txt`. All generic numbers use the same root-degree promise $\lceil s^{9/10} \rceil$. "PS" denotes polynomial space. A dash means this table does not extract a PS result from that packet, not that no other branch exists.

Packet	PS time	Cached time	Cached space
`3sat_round3_frontier`	63.75736	37.93011	33.62304
`ETH_Adaptive_Exact_Kernels`	55.33162	35.07061	33.99950
`eth_adaptive_schedules`	63.78117	37.59170	33.62304
`sat_adaptive_bounds`	**46.18485**	34.32528	34.02418
`sat_adaptive_frontier`	50.30469	31.25240	31.25240
`sat_adaptive_matrix_partition`	50.21043	34.32528	34.02418
`sat_adaptive_tree_certificate`	—	**30.89102**	**30.75391**
`subexponential_sat_adaptive_four`	50.39373	32.91581	31.66171
`subsat_rectangular`	53.97753	33.99389	33.99372

The four-block packet also gives 32.85635/31.46748 with matrix exponent 19/8. This does not change the identity of the strongest generic target. The older twenty packets establish sharp Hermite budgets, nearly lossless packing, factorial and full-block Cauchy modes, adaptive separators, fast matrix aggregation, and exact kernels. Those ingredients are credited as inputs to this report, not rediscovered contributions.

The specialized bounds must remain separate. At degree at most $s^{1/12}$, the newly inspected corpus includes a PS composition with coefficient $22.1981057366\ldots$, and the incomparable cached points $13.3796490663/13.2530249330$ and $14.2391881944/12.3613014576$. At degree at most $s^{0.51}$ there is a PS bound $25.1637654152\ldots$. These stronger input promises are not implied by our generic theorem. The full useful portfolio retains those branches, the density refinement, and the new generic branch.

2.3 What was checked in the strongest incoming proof

The tree-certificate packet bounds complete operation supports, includes the recursive cost of singleton caches, and uses a strict $9 \to 13 \to 7$ schedule. We independently reconstructed its finite integer certificates:

$$(r, Q, \text{tuples}) = (9, 3304, 9261), \quad (13, 7074, 35937).$$

Its root, depth-one, and tail branches are respectively 30.7539128321, 30.8910189204, and 30.5129274328; the depth-one branch sets the time bound. We found no defect in this scoped support/recurrence argument. Its inherited deformation and SAT reduction are still separate proof dependencies.

3 The additional Anthropic reference

The user supplied revision `1a6eee293b9e00cb432eef6018a21f523623a9af` of pull request 31 in `anthropics/formal-math`. The pull request was merged at 20:07:22 Pacific on October 26 as commit `16a5ce4130`. We inspected the supplied revision, rather than silently replacing it with a later branch state [8].

This materially updates the source record: the Lean development is now public. At the supplied revision, `eth/` contains 1,062 files, including 1,052 Lean files, of which 1,047 are under `proof/`. The earlier ProveIt statement that this development is unpublished, together with its earlier file count, is superseded. The closing ProveIt commit `7de9d09fec` already links the public development. The public `eth/paper.pdf` is a different revision from the older PDF filed in ProveIt; this report uses the public paper for the crossing-interface review.

The actual public proposition. `MainResults.MainTheorem` quantifies one deterministic `TM2ComputableAux Bool Bool` machine, with every stack alphabet finite, and one natural constant C, *before* quantifying k, n and the input formula. For $k \geq 1$, $n \geq 4$, at least two clauses, no repeated clause list, and every variable occurring, it counts satisfying assignments within

$$\left\lfloor 2^{C \min\{km/\sqrt{\log_2 m},\, k^2 n \log_2 \log_2 n/\sqrt{\log_2 n}\}} \right\rfloor \tag{4}$$

steps of the specified finite-alphabet machine. The count is returned in $n + 1$ bits. The input hypotheses deal explicitly with repeated clauses, unused variables, and the singular small logarithms; the separate ETH statement supplies its own front end. Clause order within a clause is part of the literal list encoding [9].

`Solution.lean` provides closed theorem terms of precisely `ETHIsFalse` and `MainTheorem`. The README describes the Mathlib/Lean 4.33.1 build, axiom printout, Comparator statement comparison, and nanoda check. Static inspection found the two expected challenge stubs using `sorry` in `Challenge.lean`, and no explicit axiom declarations in the Lean project. We did not rebuild the 1,052-file development or independently rerun nanoda in this session. Source presence and static inspection are not being reported as a completed kernel verification. The stated main theorem has no polynomial-space conjunct.

Effect on this research. The published bound has the same worst-case and fixed-density logarithmic orders as the comparison corpus. It has no matched numerical coefficient per normalized Cauchy-network wire, so our coefficient theorem cannot be called a numerical improvement over that distinct full algorithm. Its operator/graded-space method does give an important alternative route to an order improvement; Section 9 identifies precisely which part remains missing. The supplied source is now included in the comparison and in the archive's provenance.

4 The inherited interface and the new local task

4.1 Precisely which contraction problem is improved

The root input is a bipartite Boolean tensor network with at most s wires, maximum vertex degree at most $\lceil s^{9/10} \rceil$, and entries in

$$\mathcal{B}_{P_0,\ell} = (\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell), \qquad P_0, \ell \le s.$$

Entry routines and descriptions satisfy the source interface: polynomial space and $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ time, with fixed constants. The contraction routine works in a degree-$O(\log s)$ unramified extension of the coefficient ring. Descriptions carry immutable creation precisions: an old entry is always evaluated using its own creation-time representatives, then lifted to the current precision. Raising precision does not recompute a previous contraction or silently change a lift [1, 4].

An active call has M wires, a fixed number r of groups, retained valuation range u, cumulative packet exponent a, packet length $L = 2^a$, and a retained valuation $i < u$. Write $D = 2^{a-i}$. The inputs taken from the reviewed papers are the following, with their quantitative premises checked for our schedule in Section 7:

1. Nearly lossless asymmetric packet packing gives padded mass $(1 + o(1))M$ and child size at most $(2/r + o(1))M$.
2. Full-block integral Cauchy separation gives, for each directed pair of different groups, at most $(2 + o(1))M/(r^2D)$ families of each mode index $q \ge 0$, with guaranteed weights
$$w(i, q) = a + 1 - i + 4q - 2\,\mathrm{popcount}(q).$$
3. Sharp tapered Hermite recovery, with an additive precision guard, requires local precision $P \le (6 + o(1))MD/u$.
4. For every fixed collection of mode exponents, the deformed contraction factors into group values Z_g. A value of Z_g is a contraction of a specified smaller network, not a free oracle.

The source Cauchy deformation, integrality, recovery, exact group-factor identity, and packet-entry construction remain mathematical dependencies. Our proof changes how the resulting finite graphical sum is evaluated and how the costs are propagated. It is a theorem relative to these explicit interfaces, not an independent proof of the primary ETH result.

4.2 Exact budgets and finite index sets

Allocate a nonnegative integer budget b_{gh} to every directed pair, with $\sum_{g\neq h} b_{gh} < P$. Let $\mathcal{I}_{gh}$ be the list of its mode exponent tuples having *exact* total guaranteed weight b_{gh}. An empty list makes that allocation contribute zero. For all other allocations, put

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \qquad g_{\{g,h\}} = h_{gh} + h_{hg}, \qquad \mathcal{K}(M,u) = \kappa \frac{M}{\sqrt{u}}, \quad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

The value to compute is the exact finite sum

$$\sum_{\xi_{gh}\in\mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}). \tag{5}$$

Because our r belongs to the fixed set $\{4,7,8,9,12\}$, there are only polynomially many budget allocations. Exact list cardinalities, enumeration, and ranking/unranking use the integer knapsack recurrence

$$N(j,b) = N(j-1,b) + N(j,b-w_j), \qquad N(0,b) = \mathbf{1}_{b=0},$$

with zero values at negative budgets. The bit length of every count is polynomial in s. Choosing a contraction plan can therefore use exact integer cardinalities; it does not need floating-point logarithms.

Lemma 4.1 (Weighted mode entropy)**.** *For each fixed nonnegative vector $z = (z_{gh})$ on directed pairs,*

$$\sum_{g\neq h} z_{gh} h_{gh} \le \left(\frac{\|z\|_2}{r} + o(1) \right) \mathcal{K}(M,u). \tag{6}$$

The error is uniform over our fixed finite support families, all exact budget allocations, and all active depths of the schedule below.

Proof. This is the inherited full-Cauchy generating-function bound [5, 2]. The relevant one-family series obeys

$$\sum_{q\ge 0} -\ln(1 - e^{-\theta(1+4q-2\,\mathrm{popcount}(q))}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

For pair e with $z_e > 0$, use parameter θ/z_e in the exponential counting bound, multiply the logarithm by z_e, and sum. The leading bound is

$$\theta P + \frac{\pi^2}{24\theta} \frac{(2+o(1))M}{r^2 D} \sum_e z_e^2.$$

Optimizing θ and using $P \le (6+o(1))MD/u$ gives $\pi\sqrt{2}M\|z\|_2/(r\sqrt{u})$ in natural logarithms. Divide by $\ln 2$. Zero weights are omitted. Our guard ensures $D \ge \lambda^6$ and the saddle has scale $\sqrt{u}/D$, so the error from the displayed series is lower order uniformly. This argument does not assume an independently asymptotic approximation for each small exact list. □

4.3 Complete operation costs, including table construction

For a vertex subset A, its table sums all internal indices and retains all indices from A to its complement. Merging disjoint sets A, B, with $O = [r] \setminus (A \cup B)$, is a rectangular matrix product. Its three logarithmic dimensions are the sums H_1, H_2, H_3 of pair entropies between the classes A, O,

A, B, and B, O. With a matrix exponent strictly below $12/5$, square blocking gives the upper exponent

$$H_1 + H_2 + H_3 - \frac{3}{5}\min(H_1, H_2, H_3) = \max_j \left(\frac{2}{5} H_j + \sum_{k \neq j} H_k \right). \tag{7}$$

The final contraction of three blocks has the same triangle cost. The trace or final inner product is no more expensive. The bound pays for intermediate block tables as well as the final multiplication.

The field-to-ring bridge is the one already proved in the contraction-tree packet [2, 6]. A coefficient-ring entry is a vector of polynomially many integer coordinates. Lift the coordinates, compute their matrix products modulo enough nonexceptional odd primes, reconstruct exact integer products by CRT, apply the ring structure constants, and reduce modulo 2^P. The required reconstruction bit lengths are polynomial in s, including the logarithm of an exponential matrix dimension. A fixed rational bilinear matrix algorithm with exponent below $12/5$ supplies constant-size identities; denominators exclude only finitely many primes. Published matrix multiplication bounds suffice for this conservative exponent [11]. We do not assume that the original quotient ring is a field or divide by two or a nilpotent in that ring.

Sequential multiplication and reconstruction use a polynomial factor times the input/output table storage. Formula (7) is at least the sum of any two H_i, so it bounds all tables in that operation. Sorting records by their index tuples and scanning them costs only polynomial factors in the record and index lengths beyond the enumerated table size. This avoids an unaccounted unit-cost random-access assumption for exponential tables.

4.4 Operation supports and the prior averaging lemma

Multiply each of the three linear forms in (7) by five. The resulting vector v, indexed by unordered group pairs, has entries 0 inside a class, 2 on the discounted cross-class, and 5 on the other two cross-classes. For a complete tree plan j, let $\mathcal{V}_j$ be its set of supports and put

$$F_j(g) = \frac{1}{5} \max_{v \in \mathcal{V}_j} v \cdot g.$$

A plan first builds three nonempty blocks by balanced binary merges and then contracts their triangle. It has $r - 3$ merges and one triangle, hence $3(r-2)$ supports. The following is inherited from the latest tree packet, and is restated to make the new geometry's role explicit.

Lemma 4.2 (Support averaging)**.** *For q fixed plans, define*

$$Q = \max_{v_j \in \mathcal{V}_j} \left\| \sum_{j=1}^{q} v_j \right\|_2^2.$$

Choosing the least-cost plan gives nonrecursive time and local storage bounded, up to polynomial factors, by

$$2^{(\beta_r + o(1))\mathcal{K}(M,u)}, \qquad \beta_r = \frac{\sqrt{2Q}}{5qr}.$$

Proof. Pointwise, $\min_j F_j(g) \le q^{-1} \sum_j F_j(g) = (5q)^{-1} \max_{v_j} (\sum_j v_j) \cdot g$. Give both orientations of an edge the weight $(\sum_j v_j)_e/(5q)$. Its squared directed norm is at most $2Q/(25q^2)$; apply Lemma 4.1. The family is fixed and finite. Every input or output table is bounded by its operation cost as just shown. This controls storage, including singleton caches, but not the time spent computing their entries; that time is charged next. □

Each singleton group has $2(r-1)$ incident directed pairs. Therefore its complete cache requires at most $2^{(\alpha_r+o(1))\mathcal{K}(M,u)}$ recursive child evaluations, where

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

The fully paid recurrence is

$$T_d \le \mathrm{poly}(s)\left[2^{(\alpha_{r_d}+o(1))K_d}T_{d+1} + 2^{(\beta_{r_d}+o(1))K_d}\right], \qquad K_d = \mathcal{K}(M_d, u_d). \tag{8}$$

Summing only the local aggregation costs would miss the first term and would not establish our result.

5 The affine-plane contraction menu

5.1 Four directions, four complete plans

Identify group $3x + y$ with $(x, y) \in \mathbb{F}_3^2$. The four maps

$$y, \qquad y - x, \qquad y - 2x, \qquad x$$

partition the points into their three fibers. These are the four parallel classes of affine lines. They give the following ordered final blocks:

Direction	Final blocks, with the order used inside each block
y	$(0,3,6) \mid (1,4,7) \mid (2,5,8)$
$y-x$	$(0,4,8) \mid (1,5,6) \mid (2,3,7)$
$y-2x$	$(0,5,7) \mid (1,3,8) \mid (2,4,6)$
x	$(0,1,2) \mid (3,4,5) \mid (6,7,8)$

Within each three-point block, first merge its last two vertices, then merge the singleton with that pair. Finally contract the three block tables. Thus every plan includes six internal merges and its final triangle. Each pair of distinct points lies on exactly one affine line, and lines of different directions meet in exactly one point. These two elementary facts control the overlap of expensive operations across the four plans.

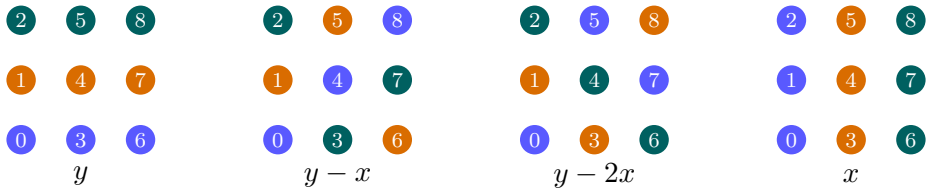


Figure 1: Each color is one final block. Coordinates and colors are interpreted modulo three. The four plans are tested on the same exact mode-list sizes.

Theorem 5.1 (Affine support certificate)**.** *For one operation support v_j chosen from each of the four affine plans,*

$$\|v_1 + v_2 + v_3 + v_4\|_2^2 \le 5400.$$

Equality is attained by every choice of four final-triangle supports. Consequently the least-cost plan has local time and local storage coefficient

$$\beta_9^2 = \frac{2 \cdot 5400}{25 \cdot 4^2 \cdot 9^2} = \frac{1}{3}. \tag{9}$$

The construction and proof allow arbitrary internal order within each line.

Proof. There are three operation types, with tripartition sizes $A = (1, 1, 7)$, $B = (1, 2, 6)$, and $C = (3, 3, 3)$. In type A the two singletons lie on the plan's affine line; in type B the first two parts form that line. For part sizes a, b, c, the largest squared support norm is

$$25(ab + ac + bc) - 21 \min(ab, ac, bc).$$

The maxima are respectively $354, 458, 486$, so $\|v_j\|^2 \leq 486$. We next prove that supports from different directions satisfy $v \cdot w \leq 288$.

Let S_v be the nonzero support and D_v the discounted subset. If

$$c = |S_v \cap S_w|, \quad a = |D_v \cap S_w|, \quad b = |D_w \cap S_v|, \quad t = |D_v \cap D_w|,$$

then

$$v \cdot w = 25c - 15(a + b) + 9t, \qquad t \leq \min(a, b). \tag{10}$$

The expression $-15(a + b) + 9 \min(a, b)$ decreases with either a or b. The following elementary counts cover all six type pairs.

For AA, each support is all edges incident to two selected collinear points. Two such pairs on transverse lines are disjoint or meet once. Their common incident edges number at most nine: the eight edges at the common point, and the edge between the other two selected points. Thus the dot product is at most 225.

For AB, enlarge the B support to all edges incident to its entire line. A transverse two-point set and three-point line have at most ten common incident edges, giving 250. For AC, the two selected points lie in different lines of the other parallel class. Each has six crossing neighbors, with their mutual edge counted twice, so $c = 11$ and the bound is 275.

For BB, two transverse underlying lines meet at x. Their full incident edge sets share twelve edges: eight at x and four between the other two points on each line. The omitted internal pair edge removes one common edge unless that operation's singled vertex is x. Unless both singled vertices are x, $c \leq 11$ and the dot product is at most 275. If both are x, the three possible discounts each meet the common support in at least two edges. Hence (10) gives at most $300 - 15(2 + 2) + 9 \cdot 2 = 258 < 275$.

For BC, the underlying three-point line meets each line of the other parallel class once. Of the fifteen crossing edges incident to it, the B support omits one, so $c = 14$. A C discount meets the B support in at least four edges. A B discount meets the C support in respectively two, four, or eight edges. Thus the dot product is at most $350 - 15(4 + 2) + 9 \cdot 2 = 278$.

For CC, the two internal-line edge sets are disjoint and each has nine edges, giving $c = 36 - 9 - 9 = 18$. Each discount has six edges crossing the other partition. The two chosen lines from each direction make a 2-by-2 grid; its two diagonals are the common discounted edges. Thus $a = b = 6, t = 2$ and

$$v \cdot w = 25 \cdot 18 - 15 \cdot 12 + 9 \cdot 2 = 288.$$

The resulting valid upper table is

	A	B	C
A	225	250	275
B	250	275	278
C	275	278	288

and all entries are at most 288. Expanding the squared sum now gives $4 \cdot 486 + 2 \cdot 6 \cdot 288 = 5400$. For four final supports, every individual squared norm is 486 and every pairwise dot product is 288, so equality holds. Lemma 4.2 proves (9). □

This proof establishes a bound for every feasible entropy vector, rather than sampling random tensor dimensions. The finite checker independently constructs all supports and verifies the norm and dot-product inequalities. A separate exhaustive check of all $21^4 = 194481$ four-tuples gives the same maximum, with exactly $3^4 = 81$ maximizers. Their squared norms sum to 622974240, an auxiliary reproducibility value.

5.2 An exact plan selector

For unordered edge $e = \{g, h\}$ define the positive integer $d_e = |\mathcal{I}_{gh}|\,|\mathcal{I}_{hg}|$. For plan j, compute

$$P_j = \max_{v \in \mathcal{V}_j} \prod_e d_e^{v_e}. \tag{11}$$

Its logarithm is $5F_j(g)$, so choosing the least P_j, with fixed-index tie breaking, chooses the least-cost plan exactly. The number of supports is fixed; the product bit lengths are polynomial. Empty lists are excluded before this selector is called. No numerical optimization is executed by the contraction algorithm.

6 The other certified menus

6.1 A twelve-group certificate with a small combinatorial core

At the second recursion level use the following four ordered plans:

Plan	Final blocks
0	$(0,1,2,3) \mid (4,5,6,7) \mid (8,9,10,11)$
1	$(9,6,4,0) \mid (1,7,10,11) \mid (5,3,2,8)$
2	$(1,2,10,4) \mid (3,11,6,9) \mid (7,0,8,5)$
3	$(0,5,10,11) \mid (3,8,1,6) \mid (4,9,7,2)$

Within each four-point block, merge its first pair and its last pair, then merge those pairs. The order in the table is part of the certificate. There are thirty supports per plan, of types $A = (1,1,10)$, $B = (2,2,8)$, and $C = (4,4,4)$.

Proposition 6.1 (Twelve-group bound)**.** *For these four plans the exact maximum squared sum of supports is $Q = 9864$. Therefore $\beta_{12}^2 = 137/400$.*

Proof. The individual squared-norm maxima are $504, 816, 864$ for A, B, C. For the six pairs of distinct plans, direct integer counts using (10) give the cross-dot upper table

	A	B	C
A	504	500	400
B	500	516	500
C	400	500	540.

This finite table can be verified from the displayed vertex sets, and the checker reconstructs all six 30-by-30 matrices. If a support tuple has an internal operation, single it out. Its squared norm is

at most 816 and each of its three cross dots is at most 516. The other three squared norms are at most 864, with mutual dots at most 540. Thus the squared norm of the sum is at most

$$816 + 3 \cdot 864 + 2(3 \cdot 516 + 3 \cdot 540) = 9744.$$

It remains to examine the three final supports in each plan. Index them by their discounted class, in order $(0,1),(0,2),(1,2)$. Their pairwise dot product is 540 when $x_j = p_{ij}(x_i)$ and 522 otherwise, where

ij	01	02	03	12	13	23
$(p_{ij}(0), p_{ij}(1), p_{ij}(2))$	$(1,2,0)$	$(1,0,2)$	$(2,0,1)$	$(0,2,1)$	$(1,2,0)$	$(2,1,0)$.

These six tiny matrices are also checked directly. Triangles $012, 023, 123$ are inconsistent: for every x, $p_{jk}(p_{ij}(x)) \neq p_{ik}(x)$. Only triangle 013 is consistent. Five edges of K_4 contain two triangles, so five preferred relations cannot hold at once. At most four pairs have dot 540, and four are attained at $(x_0, x_1, x_2, x_3) = (0,1,0,2)$. Consequently

$$Q = 4 \cdot 864 + 2(6 \cdot 522 + 4 \cdot 18) = 9864 > 9744.$$

Finally $2Q/(25 \cdot 4^2 \cdot 12^2) = 137/400$. □

The certificate was suggested by a finite search. Its validity depends only on the four displayed plans and the exact checks, not on the search or its floating-point heuristic. Independent enumeration of all $30^4 = 810000$ tuples gives nine maximizers and squared-norm sum 3501532800. No global optimality over other menus is asserted.

6.2 Puncturing the affine plane and ending the recursion

Delete point 8 from every affine block to obtain the eight-group menu. Delete points $7, 8$ for the seven-group menu. Keep the surviving order and use the same balanced split rule. No block is empty. Direct exact support certificates give

r	Plans	Supports/plan	Tuples	Q	β_r^2
9	4	21	194481	5400	1/3
12	4	30	810000	9864	137/400
8	4	18	104976	4464	279/800
7	4	15	50625	3564	891/2450
4	1	6	6	104	13/25

The four-group plan is simply $(0,1) \mid (2) \mid (3)$. Its two operations both have tripartition sizes $(2,1,1)$, up to ordering, and the largest squared support norm is 104. The punctured certificates are finite integer exhaustions, not extrapolations of the nine-point geometry. The independently reconstructed sums for $r = 8, 7$ are respectively 289547136 and 118219500.

The compact certificate theorem improves the menu coefficients within the current mode-entropy model. It does not assert that every algebraic entropy vector is realized by a SAT instance, that the chosen tree is globally optimal, or that tensor contraction requires explicit tables in every possible representation.

7 A fully paid rational recursion with five group counts

This section assumes the reviewed normalized contraction interface, rather than claiming an independent proof of the upstream SAT reduction. Let s be the root wire count, $\mathcal{L}$ the input-description length, and $\lambda = \lceil \log_2 s \rceil$. The root maximum degree is at most $\lceil s^{9/10} \rceil$; initial precision and truncation order are at most s. The imported entry-evaluation, creation-precision, padding, exact Cauchy-mode factorization, and Hermite recovery contracts are unchanged.

For the group counts used below, the certified local aggregation coefficients are

$$b_9^2 = \frac{1}{3}, \quad b_{12}^2 = \frac{137}{400}, \quad b_8^2 = \frac{279}{800}, \quad b_7^2 = \frac{891}{2450}, \quad b_4^2 = \frac{13}{25}.$$

The first is the four-direction affine-plane certificate. The next is the fixed four-plan twelve-group certificate; the eight- and seven-group certificates are punctured affine portfolios. The final coefficient uses one tree with final blocks $(0,1) \mid (2) \mid (3)$: its largest squared undirected support norm is 104, hence $b_4^2 = 2(104)/(25 \cdot 4^2)$. Each coefficient simultaneously bounds local aggregation work and local storage, including the singleton factor tables. No matrix algorithm beyond the already imported $\omega < 12/5$ bridge is needed.

7.1 All recursive factor calls are charged

Put

$$\kappa = \frac{\pi\sqrt{2}}{\ln 2}, \qquad K_d = \kappa \frac{M_d}{\sqrt{u_d}}, \qquad \alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

Each singleton factor is incident with $2(r-1)$ directed pair indices. Applying the inherited weighted pair-entropy inequality to this indicator vector bounds its number of entries by $2^{(\alpha_r + o(1))K_d}$. Every entry invokes the actual child contraction. Enumerating all exact pair-budget arrays has only polynomial cost because $r \le 12$ is fixed and all budgets are polynomial in s. Consequently the fully paid time recurrence is

$$T_d \le \mathrm{poly}(s) \left(2^{(\alpha_{r_d} + o(1))K_d} T_{d+1} + 2^{(b_{r_d} + o(1))K_d} \right). \tag{12}$$

The local storage bound is a polynomial factor times $2^{(b_{r_d} + o(1))K_d}$. Uniform input access contributes one fixed polynomial factor in $s + \mathcal{L}$ to the full algorithm: count arithmetic and entry requests first, and then charge their uniform description-access cost. It is not necessary to multiply an input-description polynomial anew at each recursion level.

7.2 The strict rational schedule

Use the group sequence

$$(r_0, r_1, r_2, r_3, r_4, r_5, \ldots) = (9, 12, 8, 7, 4, 4, \ldots)$$

and taper fractions

$$\eta_0 = \frac{63}{4000}, \qquad \eta_1 = \frac{97}{8000}, \qquad \eta_2 = \frac{21}{4000}, \qquad \eta_3 = \frac{31}{12500},$$
$$\eta_d = \frac{106449}{20000000} \left(\frac{63}{100} \right)^{d-4} \quad (d \ge 4).$$

Let

$$u_d = \lfloor \eta_d \lambda \rfloor, \qquad B = \lceil 6 \log_2 \lambda \rceil, \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)B, \quad L_d = 2^{a_d}, \quad Q_d = r_d^{a_d}.$$

The tail fractions sum to 2877/200000, and

$$\sum_{d \geq 0} \eta_d = \frac{4999}{100000} < \frac{1}{20}. \tag{13}$$

Use direct contraction when $M \leq H = \lceil s/\lambda^{3/4} \rceil$. A finite exceptional range in s can also be handled directly. Choose padding slack λ^{-3}, as allowed by the unchanged packing interface.

7.3 Packing gates and the depth bootstrap

At the root, a sufficient gate is $Q_0^2 \Delta / M = o(1)$, up to fixed polylogarithmic factors. Below the root the sufficient gates are

$$Q_d R/M = o(1), \qquad Q_d^2 C \operatorname{polylog}(s)/M = o(1), \qquad L_d Q_d^2/M = o(1),$$

where $R \leq \Delta \leq s^{9/10+o(1)}$ and $C \leq 2L_{d-1}$. The latter bound is the imported retained-side/fresh-side invariant, not an assumption that all descendants have bounded degree.

The exact integer inequalities

$$9^{100} < 2^{317}, \quad 12^{200} < 2^{717}, \quad 7^{5000} < 2^{14037}$$

give the required rational logarithm upper bounds. Using them, the gaps below exponent one in the dominant gates are

stage	0	1	2	3	$d \geq 4$
gap	29/200000	109/1600000	1/1600	42523/1000000000	1/50000.

The root row uses $2\eta_0 \log_2 9 + 9/10$; the other rows use the cumulative retained fraction times $\log_2 r_d$, plus 9/10. For the fresh-side and packet gates it suffices uniformly to use

$$\left(1 + 2 \cdot \frac{717}{200}\right) \frac{4999}{100000} = \frac{4084183}{10000000} < 1.$$

The old packet side contributes no larger exponent than the total retained fraction. Every fresh degree is at most $s^{4999/100000+o(1)}$, so it remains below Δ.

Here is a noncircular depth argument. Consider hypothetical active depths up to $\lceil (3/4) \log_{20/11} \lambda \rceil + O(1)$ and suppose $M > H$. Because the tail ratio 63/100 exceeds 11/20, its taper is at least a fixed positive constant times $\lambda^{1/4}$ throughout this range. The guard contributes $O((\log \lambda)^2)$ to every a_d. The strictly positive gaps above absorb these guards, all packing polylogarithms, and $s/M \leq \lambda^{3/4}$ uniformly. Thus padding has relative error $O(\lambda^{-3}) + s^{-c+o(1)}$ for some fixed $c > 0$. Every child has size $(2/r_d + o(1))M \leq (11/20)M$ eventually. An active path therefore cannot reach beyond the hypothetical range. This closes the depth induction and establishes that all active tapers diverge. Their floors have relative $o(1)$ error.

The product of padding errors over $O(\log \lambda)$ levels is $1 + o(1)$. Hence the child-size bounds are

$$M_0 \leq s, \quad M_1 \leq (1+o(1))\frac{2s}{9}, \quad M_2 \leq (1+o(1))\frac{s}{27}, \quad M_3 \leq (1+o(1))\frac{s}{108},$$

and

$$M_{4+j} \leq (1+o(1))\frac{s}{378\, 2^j} \qquad (j \geq 0).$$

7.4 Precision and entry costs

For retained valuations $i < u_d$ the deformation scale obeys $D = 2^{a_d - i} \geq 2^{a_{d-1}+B+1}$, with the evident root convention. The imported creation-time precision bound $p = O(s\lambda^2 2^{a_{d-1}})$ therefore satisfies

$$\frac{p}{MD/u_d} = O(\lambda^{15/4} 2^{-B}) = o(1).$$

The remaining additive Hermite loss is lower order for the same reason. Thus the sharp local evaluation precision $p_t \leq (6 + o(1))MD/u_d$ and the weighted entropy estimates hold uniformly. All ring ranks, coordinate bit lengths, and exact budgets are polynomial in s, since $a_d \leq (4999/100000)\lambda + O((\log \lambda)^2)$.

Direct leaf work is $2^{O(s/\lambda^{3/4})}$. Original entry routines cost $2^{O(s^{9/10})}$ times a polynomial, and fresh entry routines cost at most $\exp(s^{4999/100000+o(1)})$. Each is lower order on the scale $s/\sqrt{\lambda}$. Creation-time tensor descriptions remain immutable; no earlier contraction algorithm is recursively expanded inside a fresh entry routine. The multiplicities of all entry and child calls are already included in (12).

7.5 Exact time and space constants

Write

$$(m_0, m_1, m_2, m_3, m_4) = \left(1, \frac{2}{9}, \frac{1}{27}, \frac{1}{108}, \frac{1}{378}\right), \qquad K_i^* = \frac{\kappa m_i}{\sqrt{\eta_i}} \quad (0 \leq i \leq 4).$$

Successive tail coefficients have ratio

$$\gamma = \frac{1/2}{\sqrt{63/100}} = \frac{5}{\sqrt{63}} < 1.$$

Let $P_j = \sum_{i<j} \alpha_{r_i} K_i^*$. Unrolling the paid recurrence gives four finite-prefix candidates $P_j + b_{r_j} K_j^*$ for $0 \leq j \leq 3$, and the whole tail is bounded by

$$C_\infty = P_4 + K_4^* \max\left\{b_4, \frac{\alpha_4}{1-\gamma}\right\}.$$

Indeed a tail aggregation at offset t has exponent coefficient

$$P_4 + K_4^* \left(\frac{\alpha_4}{1-\gamma}(1-\gamma^t) + b_4\gamma^t\right),$$

a convex combination of the two endpoint coefficients. Here $\alpha_4/(1-\gamma) > b_4$, so the infinite leaf-route sum bounds every finite tail operation.

Exact rational outward intervals give the following certified values:

branch	coefficient, shown to 12 decimal places
0	29.487537286921
1	30.269872910441
2	29.690498000478
3	30.006711339131
∞	30.262408169228.

The second row dominates. Thus

$$C_T = \frac{\pi}{\ln 2}\left(\sqrt{\frac{128000}{5103}} + \sqrt{\frac{21920}{7857}}\right) = 30.269872910441123\ldots < 30.27. \tag{14}$$

Fixed polynomial overheads across $O(\log \lambda)$ depths add only $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$ to the logarithmic running time.

Ancestor tables coexist by addition of storage sizes, rather than by multiplication. The local memory coefficients at the five first occurrences are respectively

$$\begin{matrix} 29.487537286921\ldots & 7.570356123317\ldots & 1.934875839854\ldots \\ 0.718696654792\ldots & 0.167607514679\ldots \end{matrix};$$

later ones decrease geometrically. Therefore

$$C_S = \frac{\pi}{\ln 2}\sqrt{\frac{8000}{189}} = 29.487537286921217\ldots < 29.488. \tag{15}$$

The resulting normalized contraction has bit time

$$(s+\mathcal{L})^{O(1)} 2^{(C_T+o(1))s/\sqrt{\log_2 s}}$$

and bit space

$$(s+\mathcal{L})^{O(1)} 2^{(C_S+o(1))s/\sqrt{\log_2 s}}.$$

This improves the explicit normalized upper-bound pair $30.891018920366\ldots/30.753912832055\ldots$ from the newest prior three-plan certificate. It uses subexponential space. It neither changes the logarithmic order of the exponent nor identifies the wire count s with the number of SAT variables.

The accompanying standard-library checker proves rational logarithm gates by integer powers, encloses π by Machin's alternating series and $\ln 2$ by its positive atanh series, encloses all square roots by integer square roots, and rejects the unsupported time coefficient 30.26 and two invalid precision schedules. It checks a finite arithmetic certificate; it is not a formal proof of the imported interfaces or a full SAT solver.

8 An exact Toeplitz kernel for packet actions

This section gives a second, independent improvement: a faster exact *matrix–vector subroutine*. It does not change the entropy coefficient or the logarithmic order of the SAT bound. Toeplitz multiplication and Kronecker substitution are classical; the contribution relative to the reviewed corpus is the ring-safe packet specialization, its implementation, and its comparison with an appropriate Boolean-input control.

8.1 The convolution identity

In the source packet deformation [1], let $L = 2^a$, $S = \{-L, \ldots, L-1\}$, $\alpha = 2L$, and

$$C(T) = \prod_{q\in S}(T-q), \qquad T = t + \alpha\zeta, \quad t \in S.$$

For $-L < d < L$, define the cancelled polynomial

$$w_d = \frac{\prod_{q\in S\setminus\{d\}}(T-q)}{C'(0)}.$$

The inherited integrality statement makes these entries meaningful in the source coefficient ring. The packet matrix satisfies $W_{xy} = w_{y-x}$ for $0 \le x, y < L$ and is therefore Toeplitz. Put

$$A(X) = \sum_{d=-L+1}^{L-1} w_d X^{d+L-1}, \qquad B(X) = \sum_{y=0}^{L-1} v_y X^{L-1-y}.$$

Then the exact identity

$$(Wv)_x = [X^{2L-2-x}]A(X)B(X) \tag{16}$$

holds over every commutative coefficient ring. Zero-fill excluded columns and discard excluded rows to obtain a masked action. For rows $r + hj$ and columns $c + hk$, use the shorter Toeplitz kernel $w_{c-r+h(k-j)}$. In particular, opposite parity masks use dimension $L/2$. No Fourier transform, inverse of L, or division by a nonunit is required.

8.2 Integral diagonal preparation

The delivered implementation fixes

$$R_p = (\mathbb{Z}/2^p\mathbb{Z})[U]/(U^2 + U + 1).$$

It accepts exact integer coordinates of ζ before reducing the output. Since $U^2 + U + 1$ is irreducible modulo two, an element is a unit precisely when its two integer coordinates are not both even. Its valuation is the minimum of their two-adic valuations.

For $d \neq t$, $v_2(T - d) = v_2(t - d) \leq a$. If $\zeta \neq 0$, the remaining factor has valuation $a + 1 + v_2(\zeta)$. The complete residue grid gives

$$\sum_{q \in S \setminus \{t\}} v_2(t - q) = v_2(C'(0)), \qquad v_2(C(T)/C'(0)) = a + 1 + v_2(\zeta).$$

Cancelling any one factor leaves a nonnegative valuation. Split each exact factor into a power of two and a unit before modular reduction. Multiply the units, cancel the odd part of $C'(0)$, and batch-invert all factor units. Restore only the nonnegative final power of two. The inverse identity is

$$(a + bU)^{-1} = \frac{(a - b) - bU}{a^2 - ab + b^2};$$

the denominator is odd for a unit. Preparation uses $O(L)$ ring operations and two unit inversions. Arbitrarily long input coordinates incur their own reading and normalization cost.

If $\zeta = 0$, cancel the unique zero factor symbolically: only w_t can survive, and it is $C'(t)/C'(0)$. If $t = -L$, that diagonal is outside the matrix and the action is zero. These cases are explicitly implemented.

8.3 Exact integer convolution and bit cost

For nonnegative coordinate arrays $(a_i), (b_j)$, every ordinary convolution coefficient is at most $\min(\#a, \#b)(\max a_i)(\max b_j)$. Choose a byte-aligned power of two Q strictly larger than this bound, pack the arrays as $\sum_i a_i Q^i$ and $\sum_j b_j Q^j$, multiply integers, and unpack. No digit can carry into its neighbor. Reduce the extracted coefficients modulo 2^p. Byte-array packing avoids repeated large-integer shifts.

For coordinate arrays $a + bU$ and $c + dU$, three scalar convolutions suffice:

$$A = a * c, \quad B = b * d, \quad C = (a + b) * (c + d), \qquad (a + bU) * (c + dU) = (A - B) + (C - A - 2B)U.$$

A Boolean input uses just the two coordinate convolutions with that input. Let $M_{\mathrm{int}}(b)$ denote the bit cost of multiplying b-bit integers, and $I(p)$ that of a unit inversion at precision p. After input normalization, a conservative bound for preparation plus application is

$$O(LM_{\mathrm{int}}(p + \log L) + I(p) + M_{\mathrm{int}}(L(p + \log L))). \tag{17}$$

This includes reading and writing the full vector. Working storage for the action is $O(L(p + \log L))$ bits, in addition to whatever workspace the chosen integer multiplier needs. With a quasi-linear, near-linear-space integer multiplier the action is quasi-linear in the packed input size. The Python prototype uses its host's big-integer multiplication; it does not implement an asymptotically optimal multiplier.

The Toeplitz identity also holds for unramified extension degree e and nilpotent length ℓ. Because W has no V dependence, a conservative coordinate extension needs $e^2\ell$ scalar convolutions plus polynomial coefficient reduction. That extension is an algorithmic observation, not an implemented arbitrary-ring interface in this archive.

8.4 Matched controls, including the negative result

The independent diagonal oracle forms each cancelled product at extra precision and only then removes the known power of two. It shares neither the batch-inversion preparation nor the convolution algorithm. The action oracle performs literal matrix–vector summation. For Boolean input, a stronger control skips zero columns and masked rows and uses additions only. This matters: multiplication-heavy dense controls overstate a speedup for the actual Boolean special case.

Five-repeat synthetic timings are retained in `results/benchmarks.json`. Ratios below are control time divided by new time; a ratio below one means the new routine is slower. The general column includes preparation on both sides. Boolean columns measure action only with the same prepared diagonals.

L	p	General, incl. preparation	Boolean full grid	Boolean strided
32	32	1.656	0.178	0.332
64	64	2.432	0.208	0.427
128	64	4.123	0.324	0.691
256	128	4.434	0.345	0.846
512	128	6.643	0.491	1.272
1024	128	—	0.772	2.013

Thus the generic convolution loses to the stronger Boolean control on every sampled fixture. The strided specialization wins only at the two largest sampled sizes. This is useful implementation evidence, not a measured SAT improvement. The coefficient theorem of the preceding section does not rely on these timings or on this kernel.

9 What would change the logarithmic order?

The geometric menu removes a genuine bottleneck in the submitted constant calculation. It does not remove the partition-growth mechanism that produces the square root of the logarithm. It is useful to identify the remaining task precisely, and to separate it from optimizations that can only improve constants under the present interface.

9.1 The alternative crossing interface in the public paper

Anthropic's Appendix A gives a general crossing-piece construction with a parameter ρ. A polynomial-size family of exact crossing pieces whose row and column factorization bit counts sum to

$$4\rho\lambda + O(\log \lambda)$$

gives logarithmic denominator power $\log_2(1/\rho)$. Appendix B realizes $\rho = 1/\sqrt{2}$ using a graded Heisenberg representation and ordinary partition growth [10]. The full interface includes exact arithmetic size, piece construction, entry evaluation, and an adaptive choice of the cheaper row or column aggregation.

Appendix H already identifies the following research direction: a compatible representation with

$$\log \dim V_E \asymp E^{\alpha}$$

would suggest $\rho = 2^{\alpha-1}$ and hence logarithmic denominator power $1-\alpha$. This observation belongs to that source. A representation with $\alpha < 1/2$ would be a promising route to an order improvement *if* it also supplied all exact operator identities, locality, coefficient-size bounds, and effective evaluation guarantees. A small graded dimension by itself does not supply those properties.

The most useful research target after this review is therefore an exact, efficiently evaluable two-sided crossing decomposition with $\rho < 1/\sqrt{2}$, or an equally explicit replacement for the Cauchy weighted-mode interface. We do not construct either object here.

9.2 Why merely keeping square-indexed oscillators fails

The reviewed earlier second-round note already warned that assigning square weights to the oscillator modes breaks the necessary degree closure [7]. A different tempting change is to delete all nonsquare modes. It does improve a bare dimension count, but has a simple locality obstruction.

Let $p_{\square}(E)$ be the number of partitions of E into square parts. Then

$$\log p_{\square}(E) = \Theta(E^{1/3}).$$

For a lower bound, choose arbitrary subsets of $2^2, 3^2, \dots, J^2$ with $J = cE^{1/3}$ for a sufficiently small fixed $c > 0$, and fill the remaining weight with ones. For an upper bound, use the generating function and the Gaussian sum:

$$\begin{aligned}\log p_{\square}(E) &\le \theta E + \sum_{k\ge 1} \frac{1}{k} \sum_{j\ge 1} e^{-\theta k j^2} \\ &\le \theta E + O(\theta^{-1/2});\end{aligned}$$

take $\theta = E^{-2/3}$. This numerical growth improvement is not enough.

Proposition 9.1 (A scoped finite-pole obstruction)**.** *Let S be an infinite set of positive integers with arbitrarily long gaps, and put $a_m = m$ when $|m| \in S$ and $a_m = 0$ otherwise. The restricted scalar Heisenberg-current commutator*

$$C_S(z,w) = \sum_{m\in\mathbb{Z}} a_m z^{-m-1} w^{m-1}$$

is not annihilated by $(z-w)^H$ for any finite nonnegative integer H.

Proof. If that product vanished, coefficient comparison would give

$$\sum_{j=0}^{H} (-1)^j \binom{H}{j} a_{n-j} = 0 \quad \text{for every integer } n.$$

For $H > 0$, H consecutive zero values would force all later values to be zero by this recurrence. An arbitrarily long gap followed by another element of S is a contradiction. The case $H = 0$ is immediate. For square indices, there is the explicit witness $n = (H+1)^2$: all preceding H indices are nonsquares, and the displayed coefficient equals $n \neq 0$. □

This blocks the naive deletion of oscillator modes in the current used by the source construction. It is not a lower bound against all vertex algebras, all sparse gradings, all crossing decompositions, or all SAT algorithms. The included finite witness program is a check on the formula; the proposition itself is an all-size proof.

9.3 Further constant improvements and their limit

The support-averaging problem has an independent finite-geometric form. One seeks menus for which the sums of all possible operation supports have small Euclidean norm. The affine-plane construction works because different parallel classes are transverse, and this controls every intermediate support as well as the final triangle. It is reasonable to search other designs or weighted mixtures of plans. However, the coefficient we prove is a valid certificate for the supplied menu, not a proof that the menu or the rational schedule is globally optimal.

A pointwise cheapest-plan bound can also be strictly better than the uniform arithmetic average used by Lemma 4.2. Establishing that improvement would require a new uniform bound on $\min_j \max_{v \in \mathcal{V}_j} v \cdot g$ over the admissible nonnegative entropy vectors. Numerical optimization without a rigorous certificate would not justify a stronger constant. Likewise, letting the group count grow with input size requires controlling the number of budget allocations, all pair directions, packet packing, and table construction uniformly; our fixed-r proof does not do so.

The twelve-group menu was found by finite search and then certified independently. Its validity depends only on the frozen plans and the integer certificate. Replaying the search is not necessary to verify the result. No negative search result in this report is promoted to an impossibility theorem.

10 Verification, artifacts, and integration

10.1 What the certificate programs actually establish

All delivered correctness checkers use only the Python standard library. They use explicit exceptions rather than Python assertions, so their checks remain active under `python3 -O`. Normal and optimized executions produce identical JSON for every checker.

Checker	Evidence and scope
`check_portfolios.py`	Exhaustive integer support norms for all five group counts; compact geometric/twelve-group certificates; 100 exact plan selection cases; deliberate rejection of a false $Q = 9863$.
`check_schedule.py`	Integer logarithm certificates, rational outward intervals for π, $\ln 2$, square roots and every paid branch; strict packing gates; invalid taper/tail and 30.26 negative controls.
`check_affine_algebra.py`	45 fixtures, 153 complete planned contractions, 749 literal assignments, and 3,013 paid group-entry calls over $(\mathbb{Z}/64\mathbb{Z})[V]/(V^2)$.
`check_toeplitz.py`	1,048 packet cases; 12,328 independent cancelled diagonals; 2,096 general actions; 149 masked actions; 1,048 Boolean and 1,048 strided actions; 350 unrelated convolutions; 10 invalid inputs.
`check_lacunary.py`	Explicit nonzero finite-difference witnesses for pole orders 0 through 64.

The algebra test caches every singleton factor and compares the full plan with a literal graphical sum. Its oracle expands constant and linear coefficients directly rather than sharing the evaluator's ring

multiplication. Fixtures include zero divisors, nilpotents, distinct opposite orientations, singleton dimensions, empty allocations, and disconnected effective graphs. The finite implementation uses literal rectangular multiplication; the theoretical fast matrix backend is a separate inherited existence argument.

These tests corroborate the implementation and the finite certificates. They do not prove the all-size packing and recovery interfaces, run a full SAT solver, or constitute Lean verification of the new theorem. The all-size additions in this report are justified by the mathematical arguments, with the proof dependencies stated explicitly.

10.2 Replaying the package

From the extracted archive directory, run

```
python3 code/run_checks.py
python3 code/verify_manifest.py
bash build.sh
```

The first command runs every checker normally and with optimization, compares outputs, and writes a combined verification report. The second checks file integrity against the packaged manifest. The third builds `report.pdf` from the included LaTeX sources; it needs a normal TeX installation with `latexmk`, AMS packages, TikZ, and Latin Modern. Recorded benchmark timings are evidence from this session, not deterministic test expectations. To measure them again, use

```
python3 code/check_toeplitz.py --benchmark --output benchmarks_rerun.json
```

10.3 Contents and proof status

The archive includes the complete LaTeX source tree and compiled PDF; frozen contraction menus; exact certificate and algebra programs; normal and optimized JSON results; raw benchmark samples; a source review and independent audit notes; immutable source URLs and hashes; and a machine-readable `PROOF_STATUS.json`.

The source provenance identifies both repositories at exact commits and all nine incoming ZIP hashes. Primary upstream source files are not duplicated wholesale. The package is self-contained for replaying its new finite certificates and for building its PDF; reading the inherited proofs requires the referenced source snapshots.

For repository integration, this is a proposed new research packet. No remote branch, issue, pull request, or repository file was changed during this session. The latest repository documentation already records that the Anthropic formalization is public at the supplied revision. The new numerical frontier here is a separate normalized contraction theorem.

References

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[2] ProveIt incoming packet, *sat_adaptive_tree_certificate_2026-10-26.zip*, in the reviewed snapshot. Internal sources include its report, finite support certificates, and paid-schedule proof. Immutable source link.

[3] ProveIt incoming packet, *sat_adaptive_bounds_2026-10-26.zip*. Immutable source link.

[4] ProveIt filed packet, *eth_asymmetric_packet_balancing*, report and proof status. Immutable source link.

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[11] Josh Alman, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou, *More Asymmetry Yields Faster Matrix Multiplication*, arXiv:2404.16349. Only the conservative consequence $\omega < 12/5$ is used here. Immutable source link.